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Optimale Portfoliostrategien mit Optionen

Bachelor Thesis

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Zusammenfassung

Die vorliegende Arbeit repliziert und erweitert das von Faias und Santa-Clara (2017) eingeführte OOPS-Verfahren (*Optimal Option Portfolio Strategies*) unter Verwendung von S&P 500-Indexoptionen im Zeitraum April 2013 bis November 2025. Das Verfahren kombiniert einen bedingten Bootstrap auf historischen Indexrenditen mit der Maximierung erwarteten CRRA-Nutzens, um optimale Portfoliogewichte über vier europäische Optionen und einen risikofreien Zinssatz zu bestimmen. Transaktionskosten werden über eine Bid-Ask-Zerlegung direkt in die Optimierung eingebettet.

Ein methodischer Beitrag ist die Formulierung als gemischt-ganzzahliges konvexes Programm (MICP), gelöst mit dem MOSEK-Solver. Dies garantiert bei gegebenen Hyperparametern eine eindeutige, deterministische und global optimale Lösung an jedem Rebalancing-Termin. Darüber hinaus führen wir eine explizite Netto-Exposure-Schranke ($\delta_{\max} = 0,05$) ein, die die aggregierte Netto-Short-Position begrenzt.

Unter der Hauptspezifikation ($\bar{w} = 2\%$, $\delta_{\max} = 5\%$, $\gamma = 10$) erzielt die Strategie eine annualisierte Rendite von 21,28% bei einer Sharpe Ratio von 1,18 und einem Sicherheitsäquivalent von 17,04%. Die Renditeverteilung ist annähernd symmetrisch (Schiefe +0,09), im Gegensatz zu den stark linksschiefen Verteilungen einzelner Optionsstrategien. Univariate Faktorregressionen zeigen, dass die Marktüberschussrendite der einzige signifikante Erklärungsfaktor ist ($\beta = 0,259$, $p = 0,007$).

Eine Sensitivitätsanalyse über 64 Konfigurationen zeigt, dass das Sicherheitsäquivalent seinen Höchstwert bei $\bar{w} \in [2\%, 5\%]$ und $\delta_{\max} \in [5\%, 10\%]$ erreicht; jenseits von $\bar{w} = 7\%$ bricht die Performance zusammen. Die Arbeit dokumentiert zudem, dass die Bootstrap-Szenariogenerierung bei kurzen Volatilitätsfenstern ($d_{\text{vol}} = 1$) zu degenerierten Szenarien neigt. Die Positionsschranken fungieren daher nicht als gelegentliche Sicherheitsvorkehrung, sondern als bindende Architektur, die die Funktionsfähigkeit der Strategie gewährleistet.

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1 Introduction

Options on equity indices are among the most actively traded derivative contracts in the world, yet incorporating them into a portfolio allocation framework remains a difficult problem. The difficulty is twofold. First, option returns are far from normally distributed: long positions in puts, for instance, can return several hundred percent in a crash but lose 100% in the vast majority of months. Standard mean-variance analysis, which only considers the first two moments, is therefore inadequate for evaluating option strategies (Broadie et al., 2009). Second, the available history of option prices is much shorter than that of equity returns, making it hard to estimate return distributions directly from observed option data. These two features have led most of the academic literature to study simple, ad hoc option strategies rather than solving the full portfolio optimization problem.

A large body of work documents that simple short option positions earn abnormally high returns. Coval and Shumway (2001) show that selling crash-protected, delta-neutral straddles on the S&P 500 produces Sharpe ratios close to 1.0 over their sample period. Bondarenko (2014) finds that naked put writing generates consistently positive excess returns, and Driessen and Maenhout (2013) confirm these results for short-term options in both U.S. and U.K. markets. However, the returns of these strategies come with severe negative skewness and high kurtosis: a single tail event—such as the October 1987 crash or the COVID-19 sell-off in March 2020—can wipe out years of accumulated premium income. The certainty equivalent of these simple strategies is therefore often negative for a risk-averse investor, even though the Sharpe ratio appears attractive.

The question of whether these returns compensate for systematic risk or reflect genuine mispricing has been debated extensively. Broadie et al. (2009) argue that observed put returns are not inconsistent with a standard Black–Scholes economy in which there is no mispricing, once the sample size is taken into account. Jones (2006), using a nonlinear factor model, finds that a significant portion of index option returns can be attributed to priced risk factors. Constantinides et al. (2013), on the other hand, document a puzzle: index option returns are too low relative to what standard equilibrium models predict, suggesting that either the models are misspecified or that some form of limits to arbitrage prevents rational exploitation of the mispricing. More recently, Constantinides et al. (2020) extend this analysis to option portfolios and find that carefully constructed combinations of calls and puts can generate abnormal returns even after controlling for standard risk factors.

A separate strand of the literature addresses the portfolio allocation problem with options directly. Liu and Pan (2003) derive the optimal portfolio for a CRRA investor who can trade a stock, a 5% out-of-the-money put option, and

cash in a model with stochastic volatility and jumps. Their approach yields an analytic solution, but it requires a specific parametric model for the underlying dynamics and cannot easily be extended to more than one option. Driessen and Maenhout (2007) take an empirical approach, maximizing average realized utility over a portfolio of a stock, a single option strategy, and cash. They find that buying put options is suboptimal given historical data, because the high premium paid for downside protection reduces the portfolio’s expected utility. Eraker (2013) uses a mean-variance framework with a parametric stochastic volatility model to allocate across ATM straddles, OTM puts, and OTM calls, and reports an out-of-sample Sharpe ratio of approximately 1. However, his estimates are sensitive to the calibration period, a limitation shared by all parametric approaches. Constantinides et al. (2013) and Malamud (2014) study portfolios of both calls and puts, achieving high Sharpe ratios but with substantial negative skewness and excess kurtosis, which translates into low or negative certainty equivalents for risk-averse investors.

Faias and Santa-Clara (2017) introduce the Optimal Option Portfolio Strategies (OOPS) framework, which addresses several of the limitations described above. The key innovation is methodological: rather than specifying a parametric model for the underlying dynamics, OOPS uses a conditional bootstrap on historical index returns to generate scenarios for future option payoffs. This requires only current option quotes and a long history of the underlying asset—not a long history of option prices. The investor then maximizes expected CRRA utility over the simulated portfolio returns, choosing weights for four S&P 500 index options (ATM and 5% OTM puts and calls) and a risk-free asset. Transaction costs enter the optimization directly through the bid–ask spread, following (Eraker, 2013) and (Plyakha and Vilkov, 2008). In an out-of-sample evaluation from January 1996 to August 2013, the OOPS achieves a Sharpe ratio of 0.82 and a certainty equivalent of 9.94%, compared to 0.29 and -1.31% for the S&P 500. Notably, the OOPS return distribution exhibits positive skewness and moderate kurtosis, a marked improvement over the left-skewed, fat-tailed profiles typical of simple short option strategies. Factor regressions reveal that these returns load only weakly on market, volatility, and jump risk factors, leading the authors to conclude that the main source of OOPS returns is the exploitation of relative mispricing between options, rather than compensation for bearing systematic risk.

This thesis replicates the OOPS framework using S&P 500 index options from April 2013 to November 2025, a period that is largely disjoint from the original sample. Several features distinguish our implementation. First, we use Databento as the data source for option bid–ask quotes, rather than the OptionMetrics database used in the original study. Second, we place particular emphasis on transparency and replicability. The optimization is formulated as

a mixed-integer convex program solved with MOSEK (MOSEK ApS, 2024), which guarantees a globally optimal, deterministic solution at each rebalancing date. Given the stated hyperparameters and random seed, every allocation in this study can be exactly reproduced without multiple restarts or initialization choices. Third, we introduce an explicit net exposure constraint ($\delta_{\max} = 0.05$) that limits the aggregate net short option position, approximating the margin requirements an investor would face in practice. Our main specification uses position bounds of $\bar{w} = 2\%$.

The results confirm the viability of the OOPS in a market environment substantially different from the original sample period. The strategy earns a Sharpe ratio of 1.18 and a certainty equivalent of 17.04%, with an approximately symmetric return distribution. Univariate factor regressions, following Tables 7 and 8 in Faias and Santa-Clara (2017), show that the market excess return is the only significant contemporaneous explanatory variable, while no lagged variable significantly predicts returns at the 5% level. The remaining risk factors—changes in realized volatility, changes in VIX, and the implied volatility moneyiness spread—are uniformly insignificant. These findings are broadly consistent with the original study, although the stronger market loading in our specification reflects the binding net exposure constraint.

The remainder of this thesis is organized as follows. Section 2 describes the OOPS methodology, including the bootstrap procedure, the optimization problem, and the transaction cost treatment. Section 3 presents the data sources and summary statistics. Section 4 reports the out-of-sample performance, factor regression results, and a sensitivity analysis across 64 hyperparameter configurations. Section 5 concludes.

2 Methodology

This section presents the Optimal Option Portfolio Strategies (OOPS) framework introduced by Faias and Santa-Clara (2017), along with the specific implementation details and enhancements applied in this study. The methodology addresses three fundamental challenges in option portfolio optimization: non-normal return distributions, limited option price history, and substantial transaction costs.

2.1 Overview of the OOPS Framework

The OOPS algorithm combines bootstrap simulation with utility maximization to determine optimal portfolio allocations across European options and a risk-free asset. The key innovation lies in requiring only current option prices and historical underlying asset returns, rather than a long time series of option returns.

For each period, the algorithm executes the following sequence:

1. **Bootstrap Pool Construction:** Standardize historical index returns to create a resampling pool (Section 2.2)
2. **Scenario Generation:** Simulate future index levels through conditional bootstrap, scaled by realized volatility (Section 2.3)
3. **Option Payoff Calculation:** Compute returns for long and short positions in each available option under all scenarios (Section 2.4)
4. **Portfolio Optimization:** Maximize expected CRRA utility to determine optimal weights (Section 2.5)
5. **Out-of-Sample Evaluation:** Apply optimal weights to realized returns in the following period (Section 2.6)

Throughout this methodology, we adopt the following conventions: Let time be denoted by the subscript t and simulations by the superscript n with $n = 1, \dots, N$ where N is the total number of simulations. Our portfolio consists of $i = 1, \dots, I$ assets, and the superscripts L and S differentiate the long and short positions within asset i . Furthermore let p and c index put and call options or equations that are related to the respective option type. For each t let S_t be the index closing price, $K_{t,i}$ the strike price of option i , $A_{t,i}$ and $B_{t,i}$ indicate the Ask and Bid Price of option i . Lastly let $(\cdot|t)$ indicate that the value is conditional on information up to point t and $r_{f,t}$ the risk-free rate at t scaled to the span from t to $t + 1$.

2.2 Bootstrap Pool Construction

The foundation of the OOPS methodology is a pool of standardized returns derived from historical index data. This approach allows the algorithm to capture the empirical distribution of returns, including fat tails, skewness, and other non-normal features, without imposing parametric assumptions.

2.2.1 Daily and Monthly Returns

Let $\mathcal{D}$ denote the set of all trading days and $\mathcal{D}_t$ the subset of trading days that belong to period t . Therefore, we compute the daily returns as:

$$r_d = \log \left(\frac{S_d}{S_{d-1}} \right), \quad d \in \mathcal{D} \quad (1)$$

and the monthly returns as a simple aggregate, thanks to the additive properties of log returns:

$$r_t = \sum_{d \in \mathcal{D}_t} r_d \quad (2)$$

The full set of r_t , the raw returns, will be denoted as RR_t .

2.2.2 Realized Volatility

To standardize returns, we compute the realized volatility over each period as the square root of the sum of squared daily log-returns:

$$RV_t = \sqrt{\sum_{d \in \mathcal{D}_t} r_d^2} \quad (3)$$

This measure captures the quadratic variation of the index over period t and serves as our estimate of the period's volatility. Importantly, we do not annualize this volatility measure at this stage; the raw RV_t is used directly for standardization. The realized volatility estimator in Equation (3) is well-established in the financial econometrics literature (see Andersen et al., 2001).

2.2.3 Standardized Returns

The standardized return for period t is defined as the ratio of the raw return to the previous period's realized volatility:

$$x_t = \frac{r_t}{RV_{t-1}} \quad (4)$$

and the set of all standardized, or risk-adjusted returns will be denoted as SR_t . This normalization serves two purposes. First, it removes the time-varying scale of returns, creating a more homogeneous distribution suitable for bootstrap resampling. Second, it allows the simulation process to separately model the level (through drawn x values) and the scale (through current volatility estimates) of future returns.

Table 1: Comparison of Raw and Standardized Returns (1950–2025)

This table reports summary statistics for monthly raw and standardized returns of the S&P 500 index from January 1950 to December 2025. Statistics reported: Number of Observations (Obs), skewness (Skew), Fisher's kurtosis (Exc Kurt), autocorrelation of returns ($\rho_1(x)$) and squared returns ($\rho_1(x^2)$) for lag = 1, Ljung-Box test for autocorrelation of returns (Ljung and Box, 1978), and ARCH test (Engle, 1982).

Raw Returns				Standardized Returns			
	1950-2013	2013-2025	1950-2025		1950-2013	2013-2025	1950-2025
Obs	758.00	153.00	911.00	Obs	758.00	153.00	911.00
Skew	-0.66	-0.590	-0.65	Skew	-0.35	-1.01	-0.44
Exc Kurt	2.41	0.90	2.18	Exc Kurt	0.79	2.80	1.04
$\rho_1(x)$	0.05	-0.15	0.02	$\rho_1(x)$	0.03	-0.08	0.02
$\rho_1(x^2)$	0.14	0.35	0.16	$\rho_1(x^2)$	-0.08	-0.03	-0.07
Q1(x) Stat	2.24	3.45	0.47	Q1(x) Stat	0.72	0.94	0.22
Q1(x) p-val	0.13	0.06	0.49	Q1(x) p-val	0.40	0.33	0.64
ARCH(1) Stat	14.19	18.30	22.91	ARCH(1) Stat	5.20	0.14	4.74
ARCH(1) p-val	0.00	0.00	0.00	ARCH(1) p-val	0.02	0.71	0.03

Table 1 confirms the previously assumed characteristics of raw returns.

They exhibit significant ARCH effects and autocorrelation of squared returns, also known as volatility clusters. Interestingly, the sub-period in which this paper takes place (2013–2025) exhibits less broad tails, as indicated by the lower excess kurtosis, but even more significant autocorrelation of squared returns.

This fits expectations when taking the events of that period into account. It includes the China crash at the end of 2015 and the event broadly referred to as “Volmageddon” in 2018, when the VIX saw an intraday increase of 20 points, moving from a very calm regime and leading to the liquidation of several major VIX-based ETFs. However, the most influential event was the COVID-19 crash in early 2020 and the subsequent recovery. For example, using the expiration of options that define the optimization periods, the index lost 30% from February to March, followed by a 24% recovery in the following period. We will see further consequences in later sections when examining the characteristics of the options contracts used.

When looking at the results for our standardized returns in Table 1, there are several notable characteristics. Within the sample period from 2013–2025, the ARCH(1) p -value rises to 0.71, meaning we fail to reject the null hypothesis (H_0) of no ARCH effects. This indicates that the standardization process effectively removed the volatility clustering for this sub-period. However, the excess kurtosis significantly increased to 2.80 during this window. This suggests that while the clustering was filtered out, the magnitude of individual shocks relative to prior volatility was actually more extreme during this era than in the full 1950–2025 sample.¹

2.2.4 The Bootstrap Pool

At each decision point t , we construct the bootstrap pool as the set of all historical standardized returns conditional on the information known at t :

$$\mathcal{B}_t = \{x_1, x_2, \dots, x_{t-1}\} \quad (5)$$

As a result of the dynamic conditionality imposed, our bootstrap pool increases in each iteration ensuring that we take into account all possible information without violating information causality. For the first out-of-sample period (April 2013), $\mathcal{B}_t$ contains 758 monthly observations from February 1950 through March 2013. By the start of the final period (November 2025), it contains 909 observations.

¹ We also considered other approaches to standardize the returns but decided to stick with the original approach since they offered no meaningful impact on the mechanisms of the approach.

2.3 Scenario Generation

Given the bootstrap pool, we now generate scenarios for the next period's index level. The key challenge is to scale the standardized returns by an appropriate volatility measure that reflects current market conditions.

2.3.1 Volatility Estimation Windows

Following Faias and Santa-Clara (2017), we consider six different lookback windows for volatility estimation:

$$d_{\text{vol}} \in \{1, 5, 10, 20, 30, 60\} \text{ trading days} \quad (6)$$

For each window length d_{vol} , we select the most recent trading days prior to t and compute their realized volatility:

$$\sigma_t(d_{\text{vol}}) = RV_t(d_{\text{vol}}) \cdot \sqrt{\frac{21}{d_{\text{vol}}}} \quad (7)$$

The first term computes the realized volatility over the window, while the scaling factor $\sqrt{21/d_{\text{vol}}}$ normalizes to approximate monthly frequency (assuming 21 trading days per month on average). The multiple window lengths allow the algorithm to adapt to different volatility dynamics. Short windows ($d_{\text{vol}} = 1, 5, 10$) react quickly to volatility shocks but are noisier. Long windows ($d_{\text{vol}} = 20, 30, 60$) provide more stable estimates but lag regime changes. The optimal window is selected ex-ante based on maximized expected utility, as described in Section 2.5.

2.3.2 Return Simulation

To isolate the effect of the volatility scaling, we use the *same* set of bootstrap draws in each period t across all volatility windows d_{vol} . Specifically, for each simulation $n = 1, \dots, N$, we:

1. **Draw a standardized return** uniformly from the bootstrap pool:

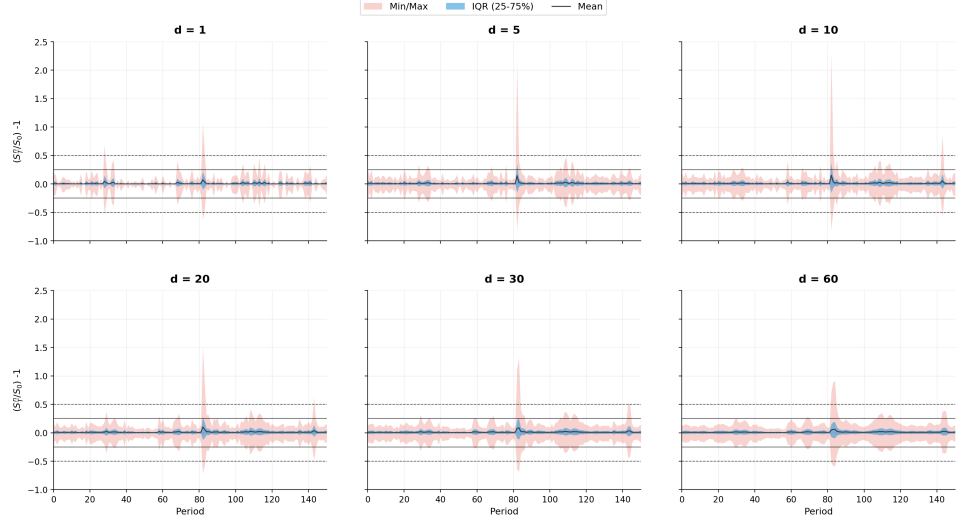
$$x^n \sim \text{Uniform}(\mathcal{B}_t) \quad (8)$$

This is sampling with replacement, so each historical standardized return has equal probability $1/|\mathcal{B}_t|$ of being selected.

2. **Scale by current volatility** to obtain a simulated period return:

$$r_{t+1|t}^n = x^n \cdot \sigma_t(d_{\text{vol}}) \quad (9)$$

Figure 1: Simulated Returns Across Volatility Windows



Time-series evolution of simulated return distributions under different realized volatility estimation windows ($d_{\text{vol}} \in \{1, 5, 10, 20, 30, 60\}$ trading days). For each period t and window length d , we generate $N = 10,000$ bootstrap scenarios and plot the distribution of normalized returns $(S^n_{t+1|t}/S_t) - 1$. The pink shaded area represents the min-max range, the blue area shows the interquartile range (25th–75th percentile), and the black line indicates the mean. The horizontal lines mark ± 0.25 and ± 0.5 for reference.

3. Generate the simulated index level at settlement:

$$S^n_{t+1|t} = S_t \cdot \exp(r^n_{t+1|t}) \quad (10)$$

The critical feature of the last equation is that it uses the current index level S_t as the starting point. This ensures that simulated scenarios are anchored to the actual market level at the decision point.

Figure 1 illustrates the tradeoff mentioned in Section 2.3 over the full sample period. The $d_{\text{vol}} = 1$ panel is particularly instructive: during calm regimes, the uncertainty bands collapse to near-zero width, implying that all simulated index levels cluster within a narrow corridor around the current spot price. As we document in Section 4.5, this compression has direct consequences for the optimization, because the optimizer systematically favors the window that maximizes expected utility—which, in calm markets, is precisely the window whose scenarios make short option positions appear riskless.

2.4 Option Payoffs and Returns

Given the simulated index levels from Section 2.3, we now calculate option payoffs and construct returns that account for transaction costs.

2.4.1 Simulated Payoffs

For each simulation n , the option payoffs at settlement are:

$$\Pi_{t+1|t,i}^{c,n} = \max(S_{t+1|t}^n - K_{t,i}, 0) \quad (11)$$

$$\Pi_{t+1|t,i}^{p,n} = \max(K_{t,i} - S_{t+1|t}^n, 0) \quad (12)$$

These payoffs represent the intrinsic value at expiration under each simulated scenario.

2.4.2 Transaction Costs via Bid-Ask Decomposition

Following Eraker (2013), Plyakha and Vilkov (2008) and therefore by extension Faias and Santa-Clara (2017), we incorporate transaction costs by decomposing each option into two securities:

Definition 2.1 (Long and Short Securities). *For each option i at time t , we define:*

- **Long security i^L :** *Purchased at the ask price $A_{t,i}$, allowing the investor to receive the payoff $\Pi_{t+1,i}$*
- **Short security i^S :** *Sold at the bid price $B_{t,i}$, obligating the investor to pay the payoff $\Pi_{t+1,i}$*

The portfolio weights $w_{t,i}^L \geq 0$ and $w_{t,i}^S \geq 0$ represent the fraction of wealth allocated to each security, subject to:

$$w_{t,i}^L \cdot w_{t,i}^S = 0 \quad (13)$$

ensuring that at most one position (long or short) is taken in each option.

This formulation embeds transaction costs directly into the optimization: going long means paying the ask, while going short means receiving only the bid. The no-short-selling constraint on each security, combined with the orthogonality constraint, prevents simultaneous long and short positions.

2.4.3 Return Calculations

The return on a long position in option i under scenario n is:

$$r_{t+1|t,i}^{L,n} = \frac{\Pi_{t+1|t,i}^n}{A_{t,i}} - 1 \quad (14)$$

This represents buying the option at the ask price $A_{t,i}$ and receiving the payoff $\Pi_{t+1|t,i}^n$.

Figure 2: Bid–Ask Spread

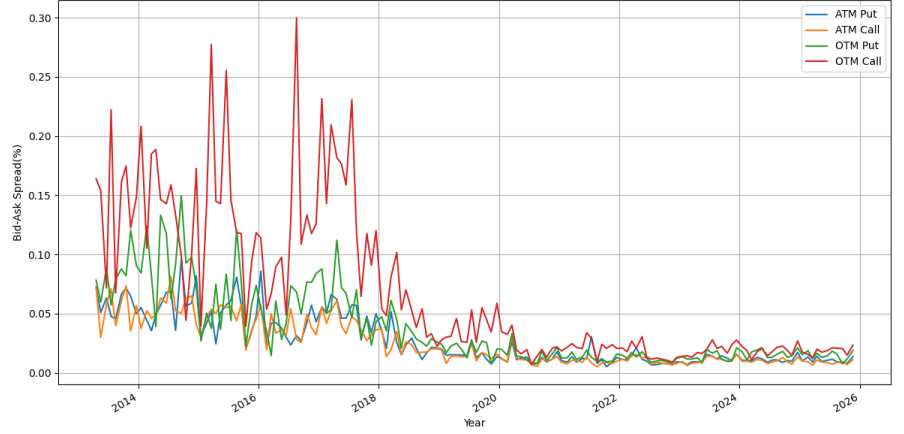


Figure 2 shows the evolution of relative bid–ask spreads from April 2013 to November 2025. The spread is defined as $\frac{\text{ask}-\text{bid}}{\text{mid}}$. Two liquidity regimes are clearly visible. From 2013 to roughly 2018/2019, relative spreads are elevated and strongly cross-sectionally dispersed, with the OTM call exhibiting pronounced spikes exceeding 25%. In contrast, after 2019 spreads compress markedly and the four series converge to consistently low levels. This pattern indicates a structural improvement in quoted option market liquidity (and thus lower transaction-cost frictions) in the later part of the sample.

The return on a short position is:

$$r_{t+1|t,i}^{S,n} = 1 - \frac{\Pi_{t+1|t,i}^n}{B_{t,i}} \quad (15)$$

This represents selling the option at the bid price $B_{t,i}$ and paying the payoff $\Pi_{t+1|t,i}^n$ at settlement.

2.5 Portfolio Optimization

With simulated returns for all options, we now construct the optimal portfolio by maximizing expected utility.

2.5.1 Portfolio Return Construction

Let $w_{t,i}^L$ and $w_{t,i}^S$ denote the weights in the long and short securities for option $i = 1, \dots, I$. The weight in the risk-free asset is implicitly determined by the budget constraint:

$$w_{t,\text{rf}} = 1 - \sum_{i=1}^I w_{t,i}^L + \sum_{i=1}^I w_{t,i}^S \quad (16)$$

The simulated portfolio return under scenario n is:

$$rp_{t+1|t}^n = w_{t,\text{rf}} \cdot r_{f,t} + \sum_{i=1}^I w_{t,i}^L \cdot r_{t+1|t,i}^{L,n} + \sum_{i=1}^I w_{t,i}^S \cdot r_{t+1|t,i}^{S,n} \quad (17)$$

where $r_{f,t}$ is the one-month Treasury bill rate scaled to the actual period length. This approach implicitly addresses margin requirements: all uninvested cash,

including premiums received from written options, is allocated to 1-month Treasury bills at the start of each period. Since Treasury holdings are generally accepted as eligible collateral, the framework reasonably approximates the margin frictions encountered in practice.

2.5.2 CRRA Utility Function

The investor's preferences are represented by a Constant Relative Risk Aversion (CRRA) utility function:

$$U_\gamma(W) = \begin{cases} \frac{W^{1-\gamma}}{1-\gamma}, & \text{if } \gamma \neq 1 \\ \ln(W), & \text{if } \gamma = 1 \end{cases} \quad (18)$$

where W is terminal wealth and $\gamma > 0$ is the coefficient of relative risk aversion. This functional form is attractive for the OOPS framework for two principal reasons. First, the homogeneity property ensures that optimal portfolio weights are independent of the initial wealth level, so maximizing $E[U(W_{t+1})]$ is equivalent to maximizing $E[U(1 + rp_{t+1})]$. Second, the concavity of the function implies that the investor cares about all moments of the return distribution, not just mean and variance. In particular, the utility function implicitly penalizes negative skewness and excess kurtosis, precisely the features that characterize option return distributions and render mean-variance optimization inadequate.

2.5.3 Optimization Problem

Given the simulated return scenarios from Section 2.3 and the portfolio return construction from Section 2.5.1, we now formulate the optimization problem that determines the optimal allocation at each rebalancing date t .

Decision Variables. For each of the $i = 1, \dots, I$ options, the investor chooses non-negative weights $w_{t,i}^L \geq 0$ and $w_{t,i}^S \geq 0$ for the long and short securities, respectively. The weight in the risk-free asset follows residually from the self-financing condition: buying long positions requires capital outlay, while selling short positions generates premium income. The budget constraint is therefore²

$$w_{t,\text{rf}} = 1 - \sum_{i=1}^I w_{t,i}^L + \sum_{i=1}^I w_{t,i}^S. \quad (19)$$

When the investor is a net seller of options, $w_{t,\text{rf}} > 1$; a net long position implies $w_{t,\text{rf}} < 1$.

² Note the asymmetry: long positions *reduce* available cash (the investor pays the ask price), while short positions *increase* it (the investor receives the bid price). This is a direct consequence of the bid-ask decomposition introduced in Definition 2.1.

Objective Function. The investor maximizes expected CRRA utility over the $n = 1, \dots, N$ simulated scenarios:

$$\max_{\{w_{t,i}^L, w_{t,i}^S\}_{i=1}^I} \frac{1}{N} \sum_{n=1}^N U_\gamma(1 + rp_{t+1|t}^n), \quad (20)$$

where γ denotes the coefficient of relative risk aversion and the simulated portfolio return $rp_{t+1|t}^n$ is defined in Equation (17).

Constraints. The optimization is subject to four groups of constraints, each serving a distinct economic and mathematical purpose. We discuss them in order of increasing complexity, because each successive layer affects the nature of the optimization problem.

(i) *Non-Bankruptcy.* Under CRRA utility with $\gamma > 1$, the function $U_\gamma(W) = W^{1-\gamma}/(1-\gamma)$ satisfies $U_\gamma(W) \rightarrow -\infty$ as $W \rightarrow 0^+$. A single scenario in which the portfolio value approaches zero would therefore dominate the entire expected utility, rendering the optimization ill-posed. We require strictly positive terminal wealth across all simulated scenarios:

$$1 + rp_{t+1|t}^n \geq \varepsilon \quad \forall n \in \{1, \dots, N\} \quad (21)$$

where $\varepsilon > 0$ is a small threshold (we use $\varepsilon = 10^{-3}$). This constraint is not merely a practical safeguard but a mathematical necessity: without it, the objective function is undefined whenever terminal wealth reaches zero. It introduces N additional linear inequalities (one per scenario), which increases the problem dimension but preserves convexity.

(ii) *Position Bounds.* Individual weights are bounded from above:

$$0 \leq w_{t,i}^L \leq \bar{w}^L, \quad 0 \leq w_{t,i}^S \leq \bar{w}^S \quad \forall i \in \{1, \dots, I\} \quad (22)$$

These linear box constraints prevent excessive concentration in a single option. Without them, the CRRA optimizer may allocate aggressively when a small number of high-payoff scenarios dominate the expected utility, a tendency amplified under finite sample sizes ($N = 10,000$), where the tails of the empirical return distribution are sparsely populated.

(iii) *Net Short Exposure Limit.* Total net short exposure is bounded:

$$\sum_{i=1}^I w_{t,i}^S - \sum_{i=1}^I w_{t,i}^L \leq \delta_{\max} \quad (23)$$

where we test several values of $\delta_{\max}$ to assess their impact on portfolio performance. This constraint serves a dual purpose: it approximates the margin requirements that an investor would face in practice, since written options

require collateral that scales with the net short position, and it acts as a regularization device that prevents the optimizer from concentrating in net short positions, a point we return to in Section 4.

Constraints (i)–(iii) are all linear in the decision variables. Together with the concave objective function, they define a *convex* optimization problem. The next constraint introduces a fundamental change.

(iv) *Mutual Exclusivity*. An investor should not simultaneously hold long and short positions in the same option, as this would generate a pure bid–ask spread loss without economic benefit. Faias and Santa-Clara (2017) describe this requirement verbally, stating that they “impose no short selling so that only one of the two, long or short positions, is ever taken” (p. 278). We enforce this mutual exclusivity explicitly as a mathematical constraint (rather than relying on numerical properties of a particular solver).

We make this requirement mathematically explicit. The underlying condition is a *complementarity constraint*:

$$w_{t,i}^L \cdot w_{t,i}^S = 0 \quad \forall i \in \{1, \dots, I\} \quad (24)$$

requiring that for each option, at least one of the two weights must be zero. This bilinear equality renders the feasible set *non-convex*: geometrically, the feasible region for each pair $(w_{t,i}^L, w_{t,i}^S)$ becomes the union of two orthogonal rays, either $w_{t,i}^L = 0$ or $w_{t,i}^S = 0$, rather than a connected set. Gradient-based solvers can no longer guarantee convergence to the global optimum. They may instead terminate at a local solution within one branch of the feasible region, without exploring the other.

Binary Reformulation. We replace the complementarity constraint with an equivalent formulation using binary variables. For each option i , we introduce binary activation indicators $x_{t,i}^L, x_{t,i}^S \in \{0, 1\}$ and impose:

$$x_{t,i}^L + x_{t,i}^S \leq 1 \quad \forall i \quad (25)$$

$$\underline{b}_i \cdot x_{t,i}^L \leq w_{t,i}^L \leq \bar{b}_i \cdot x_{t,i}^L \quad \forall i \quad (26)$$

$$\underline{b}_i \cdot x_{t,i}^S \leq w_{t,i}^S \leq \bar{b}_i \cdot x_{t,i}^S \quad \forall i \quad (27)$$

Constraint (25) ensures that at most one direction (long or short) is active per option. The weight–activation coupling (26)–(27) forces the corresponding weight to zero when the binary indicator is off ($x = 0$) and allows it to take values in $[\underline{b}_i, \bar{b}_i]$ when on ($x = 1$)³.

³ In our implementation we set the lower activation bound to zero, $\underline{b}_i = 0$, and choose the upper bound equal to the corresponding position cap (i.e., $\bar{b}_i = \bar{w}^L$ for long positions and $\bar{b}_i = \bar{w}^S$ for short positions). Hence, the binary variables only enforce the on/off (mutual exclusivity) structure and do not impose any minimum position size.

This formulation is mathematically equivalent to the complementarity constraint (24), but it transforms the problem into a *Mixed-Integer Convex Program* (MICP). The key structural property is that, for any fixed assignment of the binary variables, the remaining continuous subproblem is *convex*: the objective function remains concave and all remaining constraints are linear. This enables the MOSEK solver (MOSEK ApS, 2024) to apply a branch-and-bound algorithm with convex relaxations at each node, providing a *certificate of global optimality* upon termination.

With $I = 4$ option types, the binary reformulation introduces $2I = 8$ binary variables, yielding at most $2^8 = 256$ possible branching nodes in the worst case. In practice, the solver explores far fewer nodes than the worst-case bound due to bound-tightening and pruning.

Replicability and Solver Determinism. The MICP formulation has a practical implication that extends beyond mathematical completeness: it makes every allocation decision in this study fully replicable. Given the hyperparameters γ , $\bar{w}$, $\delta_{\max}$, and the random seed used for the bootstrap, the MOSEK branch-and-bound algorithm produces a unique, deterministic solution at each rebalancing date — no multiple restarts, no sensitivity to initial weights, and no ex-post corrections of extreme allocations are required or applied. We verify this empirically: across all 912 optimization runs (152 rebalancing periods $\times$ 6 volatility windows), MOSEK returned **OPTIMAL** status in every instance, with no **OPTIMAL_INACCURATE** solutions observed. The position bounds $\bar{w}$, the net exposure limit $\delta_{\max}$, and the risk aversion coefficient γ are therefore the sole levers through which the investor’s risk tolerance enters the strategy.

Compact Formulation. Combining all constraints, the complete optimization problem at each rebalancing date t reads:

$$\begin{aligned}
 \max_{\mathbf{w}_t, \mathbf{x}_t} \quad & \frac{1}{N} \sum_{n=1}^N U_\gamma \left(1 + w_{t,\text{rf}} \cdot r_{f,t} + \sum_i w_{t,i}^L \cdot r_{t+1|t,i}^{L,n} + \sum_i w_{t,i}^S \cdot r_{t+1|t,i}^{S,n} \right) \\
 \text{s.t.} \quad & 1 + w_{t,\text{rf}} \cdot r_{f,t} + \sum_i w_{t,i}^L r_{t+1|t,i}^{L,n} + \sum_i w_{t,i}^S r_{t+1|t,i}^{S,n} \geq \varepsilon & \forall n \\
 & 0 \leq w_{t,i}^L \leq \bar{w}^L, \quad 0 \leq w_{t,i}^S \leq \bar{w}^S & \forall i \\
 & \sum_i w_{t,i}^S - \sum_i w_{t,i}^L \leq \delta_{\max} \\
 & \underline{b}_i x_{t,i}^L \leq w_{t,i}^L \leq \bar{b}_i x_{t,i}^L, \quad \underline{b}_i x_{t,i}^S \leq w_{t,i}^S \leq \bar{b}_i x_{t,i}^S & \forall i \\
 & x_{t,i}^L + x_{t,i}^S \leq 1 & \forall i \\
 & x_{t,i}^L, x_{t,i}^S \in \{0, 1\} & \forall i
 \end{aligned} \tag{28}$$

where $w_{t,\text{rf}} = 1 - \sum_i w_{t,i}^L + \sum_i w_{t,i}^S$ is determined by the budget constraint (19).

Volatility Window Selection. The optimization in (28) is solved for each volatility window $d_{\text{vol}} \in \{1, 5, 10, 20, 30, 60\}$ separately, since each window generates a different set of simulated returns via the volatility scaling in Section 2.3. The window yielding the highest expected utility determines the final allocation for period t :

$$d_{\text{vol},t}^* = \arg \max_{d \in \{1,5,10,20,30,60\}} \frac{1}{N} \sum_{n=1}^N U_{\gamma}(1 + rp_{t+1|t}^n(d)) \quad (29)$$

However, the strict monotonicity of the CRRA function also creates a vulnerability when combined with the conditional bootstrap. If the volatility window d_{vol} is short and the recent market has been calm, the scenarios $S_{t+1|t}^n$ cluster tightly around the current spot price. In the limiting case, all simulated terminal prices fall between the OTM put and OTM call strikes, so that every OTM option expires worthless in every scenario. The optimizer then faces what appears to be a riskless opportunity: writing these options yields a return of approximately +100% of the premium in all N scenarios, producing near-maximal utility. The rational response under CRRA is to allocate the maximum permissible weight to short positions—precisely what the position bounds $\bar{w}$ and the net exposure constraint δ_{max} are designed to prevent.

This interaction has two important consequences. First, the problem is not with the CRRA function per se, but with the quality of the input scenarios: a volatility estimator that underestimates dispersion generates an artificially compressed return distribution in which tail risk is absent by construction. Second, the winner-take-all structure of the volatility window selection (Equation 29) amplifies the problem. The final allocation is determined entirely by the single d_{vol} that maximizes expected utility, with no averaging or shrinkage across windows. If one window produces degenerate scenarios while the others do not, the degenerate window will almost certainly be selected, because the absence of downside risk in its scenarios mechanically inflates its expected utility. The position constraints thus serve a dual role: they limit leverage in normal conditions, and they contain the damage when the scenario generator underestimates risk.

2.6 Out-of-Sample Evaluation

The out-of-sample evaluation follows the same steps as outlined in Sections 2.4 and 2.5, but uses realized (not simulated) payoffs.

1. We construct the actual options payoffs as:

$$\Pi_{t+1,i}^c = \max(S_{t+1} - K_{t,i}, 0) \quad (30)$$

$$\Pi_{t+1,i}^p = \max(K_{t,i} - S_{t+1}, 0) \quad (31)$$

2. The respective returns as:

$$r_{t+1,i}^S = 1 - \frac{\Pi_{t+1,i}}{B_{t,i}} \quad (32)$$

$$r_{t+1,i}^L = \frac{\Pi_{t+1,i}}{A_{t,i}} - 1 \quad (33)$$

3. Using the weights determined by the optimization, we construct the out-of-sample portfolio returns rp_{t+1} as:

$$rp_{t+1} = w_{t,\text{rf}} \cdot r_{f,t} + \sum_{i=1}^I w_{t,i}^L \cdot r_{t+1,i}^L + \sum_{i=1}^I w_{t,i}^S \cdot r_{t+1,i}^S \quad (34)$$

3 Data

As mentioned before, while traditional market data for indices such as the S&P 500 is quite readily accessible, the same does not hold for historical options data. The datasets used to model the underlying and the VIX were sourced from Yahoo Finance, or more specifically the `yfinance` library in Python (Aroussi, 2024). The data for the risk-free asset, 1-Month Treasury Bills (1M T-Bill), is directly from the FRED API, also using their Python interface (Federal Reserve Bank of St. Louis, 2024).

Due to limited public availability of historical option-level bid–ask data, we utilize the OPRA.PILLAR dataset provided by Databento (Databento, 2024). This dataset offers in-depth historical data beginning in April 2013, which therefore is also the beginning of our out-of-sample test period. To identify the relevant options, a two-phase approach, in line with the methodology outlined by Faias and Santa-Clara (2017), was used.

3.1 Options

3.1.1 Statistics

In the first phase, we used the “statistics” dataset to gain a comprehensive overview of all AM-settled S&P 500 options for each settlement date in the period. It contains all traded options and their symbols, from which we derived their expiration date⁴, strike, and type. Furthermore, it also includes the

⁴ There are a few exceptions to note for replication. First, regarding nomenclature: prior to 2015, option symbols indicated a Saturday expiration date, whereas the actual settlement occurred on the preceding Friday. Since 2015, the date within the symbol reflects the actual settlement date. Second, adjustments are necessary when the expiration date coincides with Good Friday, a US market holiday. Finally, a unique data mapping issue occurred in June 2021 due to the introduction of the Juneteenth National Independence Day. Although the holiday was observed federally on Friday, June 18, 2021, the markets remained open. However, likely due to the short notice of the holiday’s enactment, this created mapping inconsistencies where no instruments were linked to the standard expiration date of June 18, 2021. Consequently, for this specific period, the optimization was performed one day

respective open interest (OI).

The options were first grouped by the relevant moneyness buckets and ranked by their open interest. The ten instruments with the highest OI in their respective group were then selected for the next filtering step.

3.1.2 CBBO-1M

The actual price data stems from a “Consolidated Best Bid-Offer 1-Minute” (CBBO-1M) dataset, which we downloaded for all selected symbols at the relevant date. The data selection focuses on the specific interval between 16:00–16:15 Eastern Time (ET). This window captures a distinct market structure: while the primary equity markets close at 16:00 ET, trading in equity options extends until 16:15 ET. The dataset is indexed by a `ts_recv` timestamp, which records the exact minute of the order book snapshot. This granular timestamping obviates the need for the conventional practice of lagging data (e.g., the `shift(1)` method) often used to prevent look-ahead bias. Consequently, the optimization process can incorporate the full extent of the current trading day’s market data without violating information causality.

Within the resulting dataset, we choose the top-level order book bids and offers at 16:14 ET and compute the respective bid-ask spread of all instruments. At each period and for each instrument, we remove all instruments that have a bid or ask less than 0.10 and then select the instrument that offers the lowest spread within its respective moneyness bucket.

The primary resulting dataset is based on moneyness buckets as defined in Faias and Santa-Clara (2017). For the ATM options, we allow the strike to diverge up to 1% in either direction from the current spot price, while OTM options lie within 2–5% in the respective direction according to their option type. The resulting set has OTM strikes that are roughly within 2% of those described in the original paper.

This deviation from the original moneyness targets is a deliberate response to liquidity constraints in the early part of our sample. OTM calls in the early to mid 2010s were particularly illiquid: in a second dataset where we strictly enforced a 4% lower moneyness bound for OTM options, relative bid–ask spreads reached up to 120%, and in several periods fewer than ten contracts remained after filtering out instruments with bid prices below 0.10. By widening the moneyness bounds in our primary dataset, we sacrifice some distance to the original study’s strike selection in exchange for substantially better execution quality, a tradeoff we consider appropriate given the strategy’s sensitivity to transaction costs documented in Figure 2. While the strikes in this second

earlier, on June 17, 2021. To address the nomenclature discrepancies, we adjusted the code to select the closing price based on the expiration weekday (e.g., applying `shift(2)` if it is a Saturday, and `shift(1)` if it is a Thursday or Friday).

sample are now more in line with the reference, we also incur instruments with spreads up to four times wider in extreme cases compared to the main set.

The selected instruments bear similar characteristics; the divergence within Gamma and Vega is a result of the change in environment, specifically the level of the underlying index, since those Greeks are denoted in dollar value and dependent on the current spot price.

Table 2: Summary Statistics of Options (2013–2025)

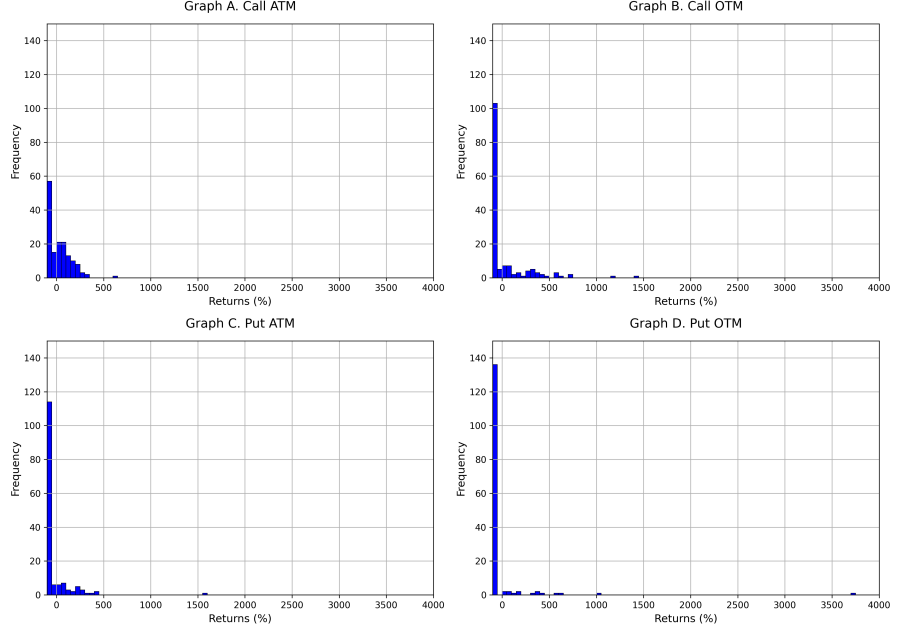
This table presents average values of key option characteristics for four contract types with approximately one month until expiration: at-the-money (ATM) calls and puts, as well as out-of-the-money (OTM) calls and puts, all evaluated at mid prices. Moneyness is measured as $S/K - 1$. The bid–ask spread denotes the absolute difference between the ask and bid quotes, while the relative bid–ask spread expresses this difference as a fraction of the mid price. Implied volatility refers to the annualized volatility extracted via the Black and Scholes (1973) pricing framework. Delta, gamma, and vega correspond to the standard Black–Scholes sensitivity measures. Elasticity is computed as the product of delta and the leverage ratio S/C (or S/P), capturing the percentage change in the option price per percentage change in the underlying.

	ATM Call	ATM Put	OTM Call	OTM Put
Option moneyness	0.44%	−0.32%	−2.40%	2.81%
Bid–ask spread	1.18	1.22	0.70	0.77
Relative bid–ask spread	2.52%	2.74%	6.79%	3.71%
Implied volatility	14.30%	14.76%	12.21%	17.58%
Delta	0.57	−0.51	0.24	−0.26
Elasticity	32.62	−31.95	60.62	−35.30
Gamma(10^2)	0.39	0.38	0.32	0.24
Vega	3.77	3.82	2.99	3.11

Comparison with Original Study. Comparing our selected options to those in Faias and Santa-Clara (2017) reveals two key differences. First, our OTM strikes are closer to at-the-money (2.4% vs. 5.0% target), which reduces crash exposure but improves liquidity. Second, relative bid–ask spreads are narrower in our sample (2.5–6.8% vs. 5–10% in the original study), likely reflecting improvements in electronic trading and market-making technology over the intervening years. These differences have implications for the strategy’s risk–return profile, as we discuss in Section 4.

Figure 3 displays the return distributions of long positions in the four option types. All four histograms confirm that option returns depart substantially from normality. The distributions are heavily concentrated near -100% , reflecting the frequent expiration at or near worthlessness, while exhibiting pronounced right tails that extend beyond $1,000\%$. This pattern is most extreme for OTM puts (Graph D), where just 15 contracts even turn a profit but within those, the median return exceeds 300% . The ATM options (Graphs A and C) show a similar but less extreme pattern, consistent with their higher intrinsic value and lower elasticity. These distributional features, extreme positive skewness, excess kurtosis, and bounded downside at -100% , render mean–variance optimization inadequate and motivate the use of expected utility maximization, which accounts for all moments of the return distribution.

Figure 3: Densities of Monthly Option Returns



Histograms of monthly raw returns of a long position in at-the-money (ATM) and out-of-the-money (OTM) options over the S&P 500 index. The period is April 2013 to November 2025.

Following Faias and Santa-Clara (2017), we evaluate the strategies using the annualized certainty equivalent of a CRRA investor with risk aversion $\gamma = 4$:

$$\text{CE} = [(1 - \gamma) \bar{U}]^{1/(1-\gamma)} - 1, \quad (35)$$

where $\bar{U} = \frac{1}{T} \sum_{t=1}^T U_{\gamma}(1 + rp_t)$. Note that this evaluation uses $\gamma = 4$, which is lower than the $\gamma = 10$ employed in the optimization. This discrepancy is intentional: the higher γ during optimization acts as a shrinkage device to prevent overfitting, while the lower γ during evaluation reflects a more realistic level of risk aversion, as estimated by Bliss and Panigirtzoglou (2004) for S&P 500 index option data.

Table 3: Summary Statistics: Levered vs. Delevered Option Strategies

	Ann. Mean	Ann. Std Dev	Min.	Max.	Skewness	Exc. Kurt.	Ann. CE(4)	Ann. Sharpe
<i>Benchmarks</i>								
S&P 500	13.31%	17.32%	-30.94%	24.71%	-1.22	13.03	6.23%	0.77
1-Month Treasury	1.70%	0.57%	0.00%	0.47%	0.85	-0.80	1.70%	
<i>Panel A: Levered Options</i>								
ATM Put	-382.34%	610.97%	-100.00%	1581.16%	5.70	45.07	-100.00%	-0.63
ATM Call	313.39%	428.54%	-100.00%	612.95%	1.16	2.19	-100.00%	0.73
OTM Put	-423.34%	1193.18%	-100.00%	3766.06%	9.29	96.86	-100.00%	-0.35
OTM Call	302.36%	885.98%	-100.00%	1554.75%	3.01	11.34	-100.00%	0.34
1/N	-47.48%	493.72%	-99.88%	1286.81%	5.49	43.39	-100.00%	-0.10
<i>Panel B: Delevered Options</i>								
ATM Put	18.86%	27.30%	-70.54%	4.96%	-5.71	45.15	-53.48%	0.69
ATM Call	-12.24%	19.15%	-27.29%	4.95%	-1.17	2.24	-18.84%	-0.64
OTM Put	20.70%	53.34%	-168.20%	4.96%	-9.29	96.81	-100.00%	0.39
OTM Call	-11.74%	39.60%	-69.49%	4.95%	-3.02	11.43	-71.95%	-0.30
1/N	7.65%	23.66%	-65.08%	4.95%	-6.38	55.32	-38.41%	0.32

Table 3 reports the performance of individual option strategies alongside two naive benchmarks. Panel A shows fully levered positions (100% of wealth in a single option), Panel B shows delevered positions in which the investor

writes options worth 4.47% of wealth, investing the resulting 104.47% cash position in the risk-free asset⁵.

The levered strategies in Panel A exhibit extreme risk characteristics. Long ATM calls generate an annualized mean return of 313% with a Sharpe ratio of 0.73, yet their certainty equivalent is -100% due to the frequent total loss of premium. ATM puts show an average return of -382% with a minimum of -100% and a maximum of $+1,581\%$. Long OTM puts display the most extreme tail behavior, with excess kurtosis of 96.86 and a maximum single-period return of 3,766%, driven by the COVID-19 crash in March 2020. Even the equal-weight ($1/N$) strategy, which allocates uniformly across all four long positions, still produces a certainty equivalent of -100% .

Panel B demonstrates that delevering alone does not resolve these problems. Individual positions retain pronounced negative skewness, ranging from -1.17 (ATM call) to -9.29 (OTM put), and substantial excess kurtosis. The delevered $1/N$ strategy yields a Sharpe ratio of 0.32 and an annualized mean return of 7.65%—not far from the S&P 500’s 13.31%—yet its certainty equivalent collapses to -38.41% , far below the benchmark’s 6.23%.

This divergence between mean return and certainty equivalent deserves emphasis. With skewness of -6.38 and excess kurtosis of 55.32, the delevered $1/N$ portfolio is exposed to rare but severe drawdowns: its worst monthly return of -65.08% would destroy nearly two-thirds of an investor’s wealth in a single period. For a CRRA investor with $\gamma = 4$, such tail events receive disproportionate weight in the utility calculation, overwhelming the moderate average gains accumulated during normal months. A mean–variance framework, which would rank the $1/N$ strategy favorably based on its Sharpe ratio, fundamentally mischaracterizes the risk–reward tradeoff when returns exhibit the extreme non-normality documented in Table 3.

These results underscore two points. First, the high Sharpe ratios of individual option strategies are misleading: once higher moments are accounted for through the certainty equivalent, most strategies destroy value for a risk-averse investor. Second, static allocation rules such as $1/N$ fail to exploit the time-varying nature of option mispricing. The OOPS algorithm addresses both shortcomings by dynamically adjusting weights across options and over time, explicitly optimizing a utility function that penalizes negative skewness and excess kurtosis.

⁵The 4.47% option exposure follows the delevering convention in Faias and Santa-Clara (2017) and is adopted here for direct comparability with their reported individual-option benchmarks.

4 Results

This section presents the out-of-sample performance of the OOPS algorithm with position bounds $\bar{w} = 2\%$ and a net exposure limit of $\delta_{\max} = 0.05$, which restricts the aggregate net short option position to at most 5% of portfolio wealth. Section 4.4 then evaluates the sensitivity of these results across 64 hyperparameter configurations.

4.1 OOPS Performance

Table 4 reports the out-of-sample performance of the OOPS alongside the S&P 500 and 1-Month Treasury benchmarks.

Table 4: OOPS Out-of-Sample Performance (April 2013 – November 2025)

	Ann. Mean	Ann. Std Dev	Min.	Max.	Skewness	Exc. Kurt.	Ann. CE(4)	Ann. Sharpe
OOPS ($\bar{w}=2\%$)	21.28%	16.56%	−15.17%	21.63%	0.09	3.24	17.04%	1.18
S&P 500	13.31%	17.32%	−30.94%	24.71%	−1.22	13.03	6.23%	0.77
1-Month Treasury	1.70%	0.57%	0.00%	0.47%	0.85	−0.80	1.70%	

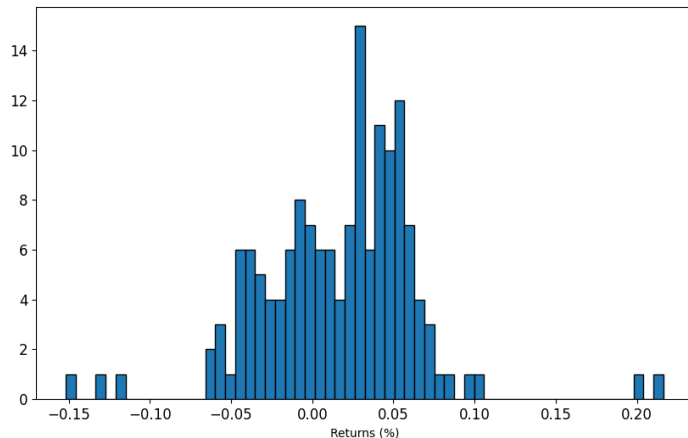
The OOPS generates an annualized mean return of 21.28% with a standard deviation of 16.56%—comparable to the S&P 500’s 17.32%—yielding a Sharpe ratio of 1.18. The annualized certainty equivalent of 17.04% nearly triples that of the S&P 500 (6.23%), confirming that the OOPS delivers substantial risk-adjusted value even after accounting for higher moments. The return distribution is approximately symmetric (skewness +0.09, excess kurtosis 3.24), and the worst monthly drawdown of −15.17% is roughly half that of the S&P 500 (−30.94%). These distributional properties are considerably more favorable than those of the individual option strategies documented in Table 3, confirming a central finding of Faias and Santa-Clara (2017): the dynamic OOPS algorithm transforms negatively skewed, leptokurtic option payoffs into a portfolio with substantially more benign characteristics.

Figure 4 illustrates the return distribution. The near-zero skewness is notable given that the underlying option positions individually exhibit extreme negative skewness. The dynamic rebalancing effectively offsets left-tail risk across the four instruments.

Figure 5 shows the cumulative wealth paths. The OOPS portfolio compounds to approximately \$12 per dollar invested (21.86% CAGR⁶), compared to roughly \$4.25 for the S&P 500 (12.1% CAGR). Two features stand out. First, the OOPS trajectory is notably smoother than the benchmark, with a maximum drawdown of −19% versus −31% for the index. Second, the COVID-19 crash in March 2020 produces a visible but contained dip in the OOPS

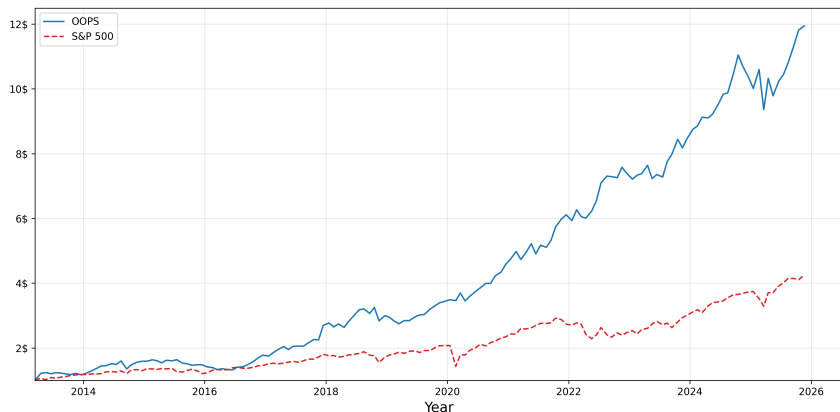
⁶The CAGR is computed from compounded monthly returns, $\text{CAGR} = (\prod_{t=1}^T (1 + r_t))^{12/T} - 1$, and differs from the annualized arithmetic mean $12\bar{r}$ because compounding is nonlinear (Jensen’s inequality); for positive average returns and sufficiently low volatility, the compounded growth rate can exceed the linear annualization.

Figure 4: OOPS Return Distribution



Histogram of monthly OOPS excess returns from April 2013 to November 2025. The distribution is approximately symmetric with low excess kurtosis, in contrast to the heavily left-skewed distributions typical of individual option strategies (Table 3).

Figure 5: OOPS Cumulative Returns



Cumulative wealth of \$1 invested in the OOPS portfolio (solid) and the S&P 500 total return index (dashed) from April 2013 to November 2025.

portfolio, after which the strategy accelerates its outperformance—suggesting that the algorithm successfully adapted its allocation during the subsequent high-volatility environment. With approximately 101.41% of portfolio wealth in Treasury bills and individual option positions of at most 2%, market impact is negligible, margin requirements are trivially satisfied, and the implementation frictions discussed in Section 4.5 are likely second-order for the reported performance.

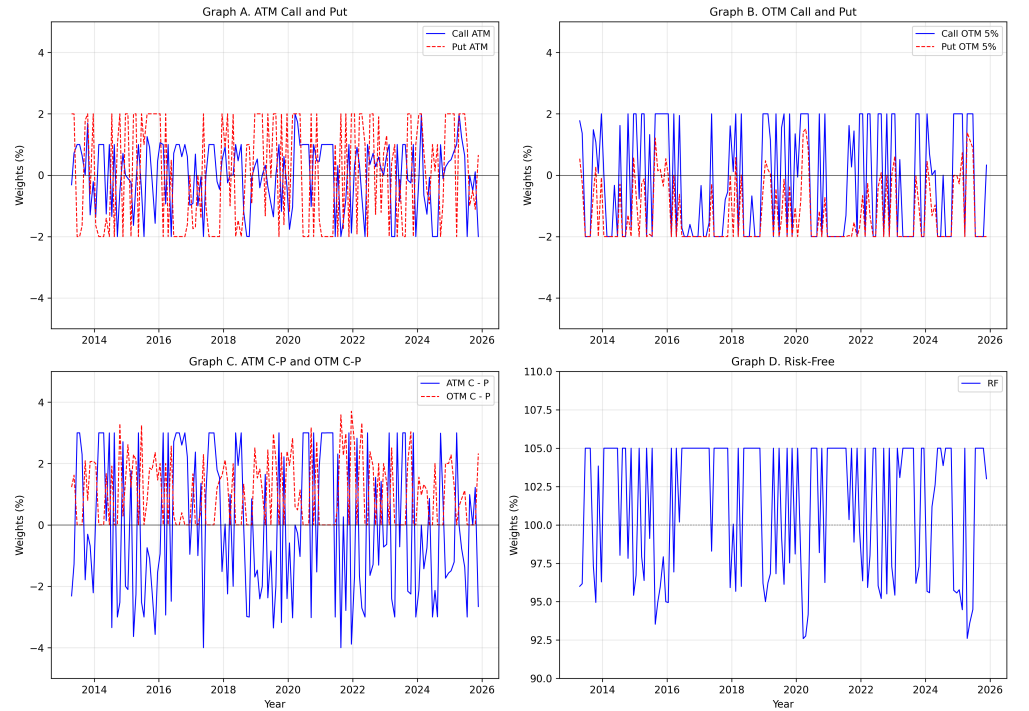
Comparison with the original study. Faias and Santa-Clara (2017) report an annualized certainty equivalent of 9.94% and a Sharpe ratio of 0.82 over January 1996 to August 2013. Our OOPS achieves higher absolute performance, but the S&P 500 itself performed substantially better in our sample period, with a Sharpe ratio of 0.77 versus 0.29 in the original. Relative to the market, the OOPS-to-benchmark Sharpe ratio is $1.5\times$ in our sample compared

to $2.8\times$ in the original study. The strategy’s relative outperformance is therefore more modest in the modern period. The higher absolute returns likely reflect two environmental factors: the prolonged bull market from 2013 to 2025, during which written options expired worthless more frequently, and the narrower bid–ask spreads documented in Table 2, which reduce the effective transaction costs of dynamic rebalancing.

4.2 Portfolio Weights

Figure 6 and Table 5 characterize the OOPS allocation over the 152-month sample.

Figure 6: OOPS Portfolio Weights Over Time ($\bar{w} = 2\%$, $\delta_{\max} = 0.05$)



Graph A and B show the net weights (long minus short, as percentage of portfolio wealth) for each of the four option types. Positive values indicate net long positions; negative values indicate net short positions. Graph C displays the call-minus-put differential within each moneyness bucket. Graph D reports the residual risk-free weight, which fluctuates between 93% and 105% and exceeds 100% whenever the portfolio is a net seller of options. The period spans April 2013 to November 2025.

Table 5: Portfolio Weight Summary Statistics ($\bar{w} = 2\%$, $\delta_{\max} = 0.05$)

Weight is defined as $w_i^L - w_i^S$ for each option i . The risk-free weight follows from constraint (19).					
	ATM Put	ATM Call	OTM Put	OTM Call	Risk-free
mean	0.07%	0.01%	-1.25%	-0.24%	101.41%
min	-2.00%	-2.00%	-2.00%	-2.00%	92.59%
max	2.00%	2.00%	1.51%	2.00%	105.00%

The algorithm actively switches between long and short positions in all four instruments, confirming that the dynamic rebalancing captures time-varying relative value opportunities across options rather than maintaining a static

directional bet. The risk-free weight averages 101.4%, consistent with the net option-selling posture reported by Faias and Santa-Clara (2017) (mean risk-free weight of 104.5%). Graph C shows that the call–put differential is predominantly positive for both ATM and OTM pairs, indicating a systematic preference for selling puts relative to calls—a pattern consistent with harvesting the well-documented put premium in index options markets. Among individual instruments, OTM puts exhibit the strongest directional tilt, with a mean net weight of -1.25% that is an order of magnitude larger than that of any other instrument, serving as the primary vehicle for premium collection. OTM calls also carry a negative mean weight (-0.24%), while the ATM instruments are approximately neutral. This persistent short bias in OTM puts is consistent with harvesting the variance risk premium embedded in index options.

4.3 Factor Regressions

Following Faias and Santa-Clara (2017), Tables 7 and 8, we run univariate regressions to investigate whether OOPS returns load on known risk factors. The explanatory regressions regress monthly excess returns on one contemporaneous variable at a time; the predictive regressions use lagged variables as predictors. All regressions employ Newey–West HAC standard errors with four lags. The contemporaneous variables are: the market excess return ($r_{\text{mkt}} - r_f$), a jump risk measure (JUMP), changes in realized volatility (ΔRV), changes in the VIX index (ΔVIX), and the implied volatility moneyness spread (IV Moneyness) defined as $\text{IV Moneyness} = \text{IV}_{\text{Put}_{\text{ATM}}} - \text{IV}_{\text{Call}_{\text{ATM}}}$. The JUMP variable follows Faias and Santa-Clara (2017): it equals the market excess return whenever $r_{\text{mkt}} - r_f < -5\%$, and zero otherwise, capturing both the occurrence and severity of market crashes. The predictive regressions add the lagged VIX level, realized volatility, variance risk premium ($\text{VRP} = \text{VIX} - \text{annualized RV}$), and portfolio gamma as predictors.

Table 6: Univariate Explanatory Regressions ($\bar{w} = 2\%$, $\delta_{\text{max}} = 0.05$)

	$r_{\text{mkt}} - r_f$	JUMP	ΔRV	ΔVIX	IV Moneyness	N
β	+0.259*** [0.007]	+0.310 [0.092]	−0.008 [0.160]	−0.020 [0.161]	−0.375 [0.313]	152
$\bar{R}^2$	6.7%	3.5%	0.7%	1.4%	0.0%	

$r_t - r_{f,t} = \alpha + \beta X_t + \varepsilon_t$. Each row is a separate regression. p -values in brackets; HAC standard errors (Newey–West, 4 lags). *** $p < 0.01$, ** $p < 0.05$.

Table 6 shows that the market excess return is the only significant contemporaneous factor ($\beta = +0.259$, $p = 0.007$, $\bar{R}^2 = 6.7\%$). The strategy participates in roughly one quarter of market movements, reflecting the combination of $\delta_{\text{max}} = 0.05$ and tight position bounds that induces a systematic long bias through binding constraints on net short positions. In Faias and Santa-Clara (2017), who impose no net exposure constraint, the market coefficient is smaller (0.14, $p = 0.04$, $\bar{R}^2 = 1.46\%$). JUMP, changes in realized

volatility, changes in VIX, and the IV Moneyness spread are all insignificant, consistent with Faias and Santa-Clara (2017), Table 7: the strategy does not load on volatility changes or the slope of the implied volatility smile.

Table 7: Univariate Predictive Regressions ($\bar{w} = 2\%$, $\delta_{\max} = 0.05$)

	RV _{<i>t</i>-1}	VIX _{<i>t</i>-1}	VRP _{<i>t</i>-1}	JUMP _{<i>t</i>-1}	IV Mon. _{<i>t</i>-1}	Gamma _{<i>t</i>-1}	<i>N</i>
β	-0.000 [0.684]	-0.001 [0.101]	-0.000 [0.429]	-0.142 [0.065]	+0.211 [0.528]	-22.17 [0.315]	151
$\bar{R}^2$	-0.6%	0.6%	-0.3%	0.3%	-0.4%	-0.1%	

$r_t - r_{f,t} = \alpha + \beta Z_{t-1} + \varepsilon_t$. Each row is a separate regression. *p*-values in brackets; HAC standard errors (Newey–West, 4 lags). One observation lost to lagging. ****p* < 0.01, ***p* < 0.05. VRP = VIX – annualized realized volatility. Gamma is the portfolio-weighted Black–Scholes gamma.

Table 7 shows that no lagged variable significantly predicts OOPS returns at the 5% level, with all $\bar{R}^2$ values near zero or negative. This is consistent with Faias and Santa-Clara (2017), where no predictor reaches significance (Table 8, maximum $\bar{R}^2 = 0.41\%$). The absence of predictability is expected for a strategy whose returns are driven by option premium collection rather than by directional forecasts of the underlying: lagged volatility measures and crash indicators carry little information about whether subsequently written options will expire in or out of the money. We also considered multivariate specifications combining several of the regressors above. However, given the high pairwise correlation among some of them, especially the volatility-related variables ($\Delta RV, \Delta VIX, VRP$), multivariate models introduced multicollinearity without improving explanatory power, and we therefore report only the univariate results.

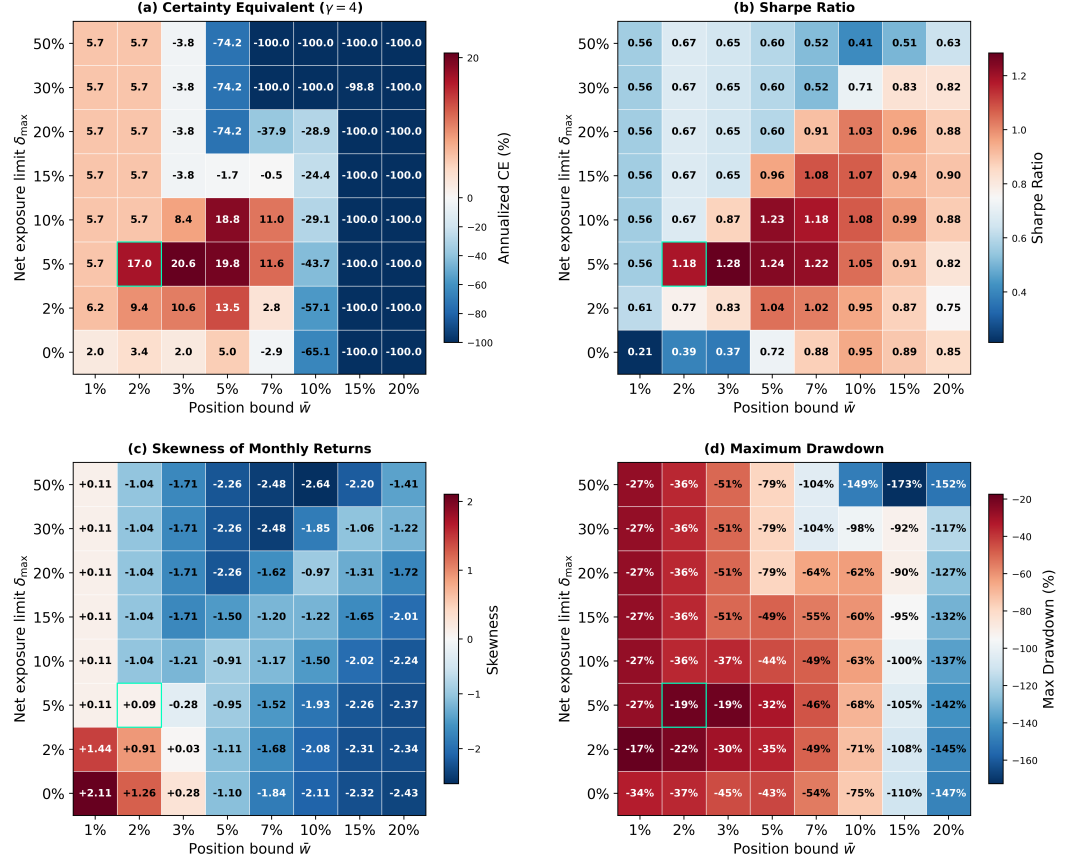
4.4 Sensitivity Analysis

The results presented so far are conditional on the specific choice of $\bar{w} = 2\%$ and $\delta_{\max} = 0.05$. To assess how sensitive the strategy’s performance is to these hyperparameters, we evaluate the OOPS over a grid of $8 \times 8 = 64$ configurations with $\bar{w} \in \{1, 2, 3, 5, 7, 10, 15, 20\}\%$ and $\delta_{\max} \in \{0, 2, 5, 10, 15, 20, 30, 50\}\%$. All other parameters—including $\gamma = 10$, the bootstrap seed, and the number of scenarios—are held fixed. Each configuration is evaluated out of sample over the full 152-month period.

Performance heatmap. Figure 7 reports the annualized certainty equivalent ($\gamma = 4$), Sharpe ratio, skewness, and maximum drawdown for each configuration.

The CE surface exhibits a sharply defined optimum in the region $\bar{w} \in [2\%, 5\%]$, $\delta_{\max} \in [5\%, 10\%]$. The global maximum is attained at ($\bar{w} = 3\%$, $\delta_{\max} = 5\%$) with CE = 20.6%, closely followed by ($\bar{w} = 5\%$, $\delta_{\max} = 5\%$) at 19.8% and ($\bar{w} = 5\%$, $\delta_{\max} = 10\%$) at 18.8%. Our primary specification at ($\bar{w} = 2\%$, $\delta_{\max} = 5\%$) achieves 17.0%, placing it squarely within this optimal zone.

Figure 7: OOPS sensitivity analysis



Panel (a): annualized certainty equivalent ($\gamma = 4$); Panel (b): annualized Sharpe ratio; Panel (c): monthly return skewness; Panel (d): maximum drawdown. Sample period: April 2013–November 2025.

Outside the sweet spot, performance deteriorates rapidly. For $\bar{w} \geq 10\%$, the CE collapses to approximately -100% regardless of $\delta_{\max}$. This reflects the CRRA utility's extreme sensitivity to tail events: a single month in which a short option position produces a loss exceeding the allocated weight generates a wealth realization near zero, and the term $(1 + r)^{1-\gamma}$ with $\gamma = 4$ amplifies this into a utility of $-\infty$ that dominates the sample average.

Panel (b) reveals that the Sharpe ratio surface is broader and more forgiving: at $(\bar{w} = 10\%, \delta_{\max} = 5\%)$, the Sharpe ratio is still 1.05, even though the CE has already fallen to -43.7% . This confirms that the Sharpe ratio is a misleading performance metric for option strategies, as it ignores the higher moments that dominate the investor's welfare. The Sharpe ratio ranking nevertheless agrees with the CE ranking within the optimal zone, with the top five cells coinciding across both panels.

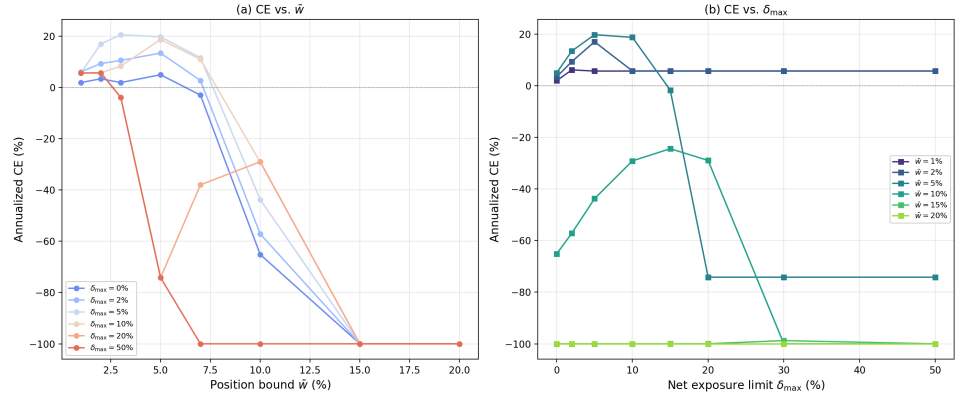
Panel (c) explains the mechanism behind the CE collapse. Only 13 of the 64 configurations exhibit positive skewness, all concentrated at $\bar{w} \leq 3\%$. At our specification $(\bar{w} = 2\%, \delta_{\max} = 5\%)$, skewness is $+0.09$; at $(\bar{w} = 3\%, \delta_{\max} = 5\%)$, it has already turned negative at -0.28 ; and at $(\bar{w} = 5\%, \delta_{\max} = 5\%)$, it reaches -0.95 . The positive skewness that distinguishes the OOPS from simple short option strategies is thus a fragile property that exists only under tight position

bounds.

Panel (d) corroborates this pattern: the maximum drawdown exceeds -50% for all configurations with $\bar{w} \geq 10\%$, whereas it remains moderate at -19.3% for our primary specification. The drawdown surface also demonstrates the non-linear interaction between $\bar{w}$ and $\delta_{\max}$: even at $\bar{w} = 5\%$, relaxing $\delta_{\max}$ from 5% to 20% increases the maximum drawdown from -31.7% to -78.6% .

Sensitivity profiles. Figure 8 traces the CE as a function of each hyperparameter while holding the other fixed at selected values.

Figure 8: CE sensitivity profiles



Panel (a): CE as a function of $\bar{w}$ for selected $\delta_{\max}$. Panel (b): CE as a function of $\delta_{\max}$ for selected $\bar{w}$. All values annualized with $\gamma = 4$.

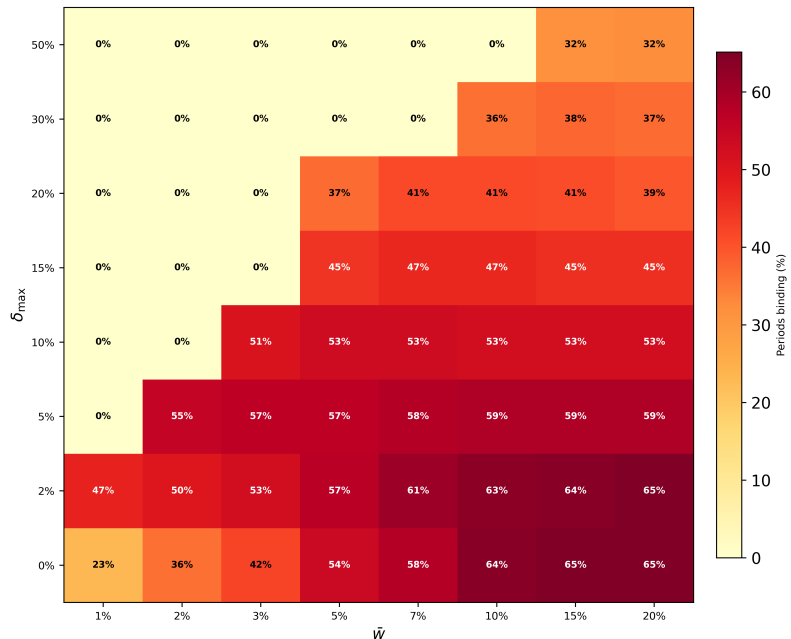
Panel (a) shows that the CE profile is hump-shaped in $\bar{w}$: for $\delta_{\max} = 5\%$, CE rises from 5.7% at $\bar{w} = 1\%$ to a peak of 20.6% at $\bar{w} = 3\%$, then declines steeply, turning negative beyond $\bar{w} = 7\%$. The ascent reflects the higher premium income from larger option positions; the descent reflects the increasing weight of tail losses in CRRA utility. At stricter $\delta_{\max} = 2\%$, the hump is flatter and the peak shifts rightward to $\bar{w} = 5\%$ (CE = 13.5%), because the tighter net exposure constraint suppresses tail risk even at higher position sizes.

Panel (b) shows that the CE profile in $\delta_{\max}$ is non-monotonic for $\bar{w} \geq 3\%$. At $\bar{w} = 5\%$, CE peaks at $\delta_{\max} = 5\%$ and declines sharply as the constraint is relaxed. For $\bar{w} = 1\%$, the curve is essentially flat above $\delta_{\max} = 5\%$, confirming that the constraint is never active when positions are small enough. Conversely, at $\bar{w} = 10\%$ and above, the CE is negative for all values of $\delta_{\max}$, and the constraint cannot rescue the strategy from excessive position sizes.

Constraint binding rate. Figure 9 reports the fraction of the 152 monthly rebalancing dates at which the $\delta_{\max}$ constraint is approximately binding.

At our primary specification ($\bar{w} = 2\%$, $\delta_{\max} = 5\%$), the constraint binds in 55% of months, confirming that it actively shapes portfolio composition rather than serving as an inactive safeguard. For tighter limits ($\delta_{\max} = 2\%$), the

Figure 9: Constraint binding rate



Fraction of periods in which the $\delta_{\max}$ constraint is binding (within 0.5 percentage points of the limit). Sample period: April 2013–November 2025.

binding rate rises to 47–65%. For looser limits ($\delta_{\max} \geq 30\%$), it falls below 38% and often to zero, indicating that the unconstrained optimizer naturally selects net short exposure within these bounds. The binding rate also increases with $\bar{w}$: at $\delta_{\max} = 5\%$, it rises from 0% at $\bar{w} = 1\%$ (where position size alone prevents the limit from being reached) to 59% at $\bar{w} \geq 10\%$ (where the optimizer consistently pushes against the constraint).

4.5 Caveats

The results presented above must be interpreted in light of several important limitations, particularly given that our findings deviate from the original study in ways that warrant careful qualification.

Hyperparameter selection. The position bound $\bar{w} = 2\%$, the net exposure limit $\delta_{\max} = 0.05$, and the risk aversion coefficient $\gamma = 10$ were selected based on economic reasoning and the calibration of the original study, but were not pre-registered. The sensitivity analysis in Section 4.4 shows that the CE optimum is sharply localized around $\bar{w} \in [2\%, 5\%]$ and $\delta_{\max} \in [5\%, 10\%]$, which mitigates the concern of data snooping to some extent—the selected specification lies within a stable plateau rather than at an isolated peak—but the possibility of implicit overfitting cannot be fully excluded. A stricter test would fix all hyperparameters ex ante and evaluate on a hold-out period, a design that the limited sample size of 152 monthly observations does not readily support.

Sample period bias. The out-of-sample period from April 2013 to November 2025 coincides with one of the longest equity bull markets in U.S. history, interrupted only briefly by the COVID-19 crash in March 2020. This environment is structurally favorable to option-writing strategies: sustained upward drift causes written puts to expire worthless with high frequency, inflating mean returns and suppressing realized crash losses. The S&P 500 achieved a Sharpe ratio of 0.77 in our sample, compared to 0.29 in the period studied by Faias and Santa-Clara (2017). It is therefore unclear to what extent the strong absolute performance of the OOPS reflects the quality of the algorithm versus the favorability of the market environment. The strategy’s behavior during a prolonged bear market, a sustained high-volatility regime, or a period of rising interest rates remains untested.

Transaction costs and implementation frictions. Our transaction cost model prices all trades at the quoted bid–ask spread: long positions are entered at the ask and short positions at the bid. While this captures the first-order cost of trading, it does not account for several frictions that would affect a real-world implementation. Market impact—the adverse price movement caused by the order itself—is not modeled. For OTM options with limited order book depth, even moderate institutional-scale orders could move prices substantially beyond the quoted spread. Similarly, the model assumes that quoted prices are executable at the reported sizes, which may not hold during periods of market stress when liquidity withdraws precisely when it is most needed. These frictions would be more severe for larger position sizes. The true implementable performance is therefore likely lower than reported, though the magnitude of the reduction is difficult to quantify without order-level data. At $\bar{w} = 2\%$, however, option positions represent negligible order flow relative to the liquidity available in S&P 500 index options, making these concerns largely irrelevant for our specification.

Volatility window selection and scenario degeneracy. The grid search over volatility windows $d_{\text{vol}} \in \{1, 5, 10, 20, 30, 60\}$ introduces a subtle but economically consequential issue at the shortest window. When $d_{\text{vol}} = 1$, the volatility estimate is derived from a single trading day’s squared return, scaled by $\sqrt{21}$. After a calm day—say $|r_d| = 0.05\%$ —the estimated monthly volatility is approximately 0.23%, implying that all $N = 10,000$ simulated index levels cluster within a band of roughly $\pm 1\%$ around the current spot price. For OTM options with strikes 2–5% away from the money, this produces scenarios in which *no* simulated path causes the option to expire in the money. From the perspective of a CRRA investor with $\gamma = 10$, the optimal response is then to write all available options at the maximum permitted weight: the simulated

premium income is certain, the simulated payoff obligation is zero, and the expected utility gain is unambiguous.

Although our objective is utility-based portfolio choice rather than VaR estimation, the underlying scenario mechanism is the same: a finite-sample, non-parametric resampling of historical (standardized) returns, which can become overly benign in low-volatility regimes. This behavior reflects a well-documented limitation of non-parametric scenario-generation methods based on historical resampling (including historical simulation and bootstrap approaches): because the scenario distribution is drawn from a finite historical window and then re-scaled by an estimated conditional volatility, tail risk can become understated in tranquil, low-volatility regimes and risk estimates adjust sluggishly to changes in conditional risk. As Pritsker (2006) shows, historical simulation methods are under-responsive to changes in conditional risk and can react asymmetrically, with measured risk rising after large losses but not comparably after large gains. The filtered historical simulation (FHS) approach of Barone-Adesi et al. (1999) partially mitigates this by filtering returns through a GARCH-type volatility model—devolatilizing returns, resampling standardized innovations, and re-scaling by the current volatility. However, its effectiveness is only as good as the conditional volatility estimate and can be noisy in small samples (Pritsker, 2006); in our setting this bottleneck is most apparent for $d_{\text{vol}} = 1$, where volatility estimation noise and occasional volatility compression can yield overly concentrated scenarios in calm regimes.

In practice, the volatility window selection mechanism does *not* reliably mitigate the problem. In the primary specification, $d_{\text{vol}} = 1$ is selected in 96 of the 152 rebalancing periods (63.2%), making it the dominant volatility window by a wide margin.⁷ This reveals that the winner-take-all structure of the volatility window selection (Equation 29) systematically favors the window that produces the narrowest scenario distribution: after a calm trading day, the single-day volatility estimate generates scenarios in which all OTM options expire worthless, making short positions appear riskless to the optimizer and thereby maximizing expected utility by construction. The position bounds and net exposure limits are therefore not a safeguard for rare edge cases but a binding constraint that actively shapes portfolio composition in the majority of periods. This finding reinforces the conclusion from Section 4.4 that the strategy’s viability is inseparable from its constraint architecture.

Statistical power and alpha significance. The univariate regressions explain at most 6.7% of the return variation (market excess return), leaving the bulk of OOPS returns unexplained by the variables we consider. Faias and Santa-Clara (2017) report similarly low $\bar{R}^2$ values and conclude that OOPS

⁷The full distribution of selected windows is: $d_{\text{vol}} = 1$: 96, $d_{\text{vol}} = 5$: 15, $d_{\text{vol}} = 10$: 13, $d_{\text{vol}} = 20$: 6, $d_{\text{vol}} = 30$: 6, $d_{\text{vol}} = 60$: 16.

returns are primarily driven by the exploitation of relative option mispricing rather than risk factor exposure. Our results are broadly consistent: the market excess return is the only significant factor ($\bar{R}^2 = 6.7\%$), with a stronger loading than in the original study, while all other variables do not reach significance at the 5% level. Whether the unexplained component reflects genuine mispricing or merely omitted risk factors cannot be resolved with 152 observations.

Divergence from the original study. Several of our findings differ quantitatively from Faias and Santa-Clara (2017). Under our tight constraints, the OOPS portfolio exhibits comparable volatility (16.6% vs. 18.4% in the original) and positive skewness (+0.09 vs. +0.35). However, the market loading is stronger ($\beta = 0.26$ vs. 0.14), reflecting the binding net exposure constraint. Part of these differences may be attributable to different market regimes: the inclusion of the COVID-19 crash—the fastest 30% drawdown in S&P 500 history—introduces a tail event of a severity not observed in the original sample. Additionally, differences in the optimization implementation (CVXPY with MOSEK vs. MATLAB FMINCON) and the option data source (Databento vs. OptionMetrics) may contribute to quantitative discrepancies. These differences do not invalidate the results, but they caution against interpreting our findings as a direct replication of the original study. Rather, they represent an independent out-of-sample evaluation of the OOPS methodology under different market conditions and with a different data infrastructure.

5 Conclusion

This thesis applies the OOPS framework of Faias and Santa-Clara (2017) to S&P 500 index options from April 2013 to November 2025.

One methodological contribution of this thesis is the deliberate design of a fully transparent and deterministic implementation. Given the hyperparameters γ , $\bar{w}$, $\delta_{\max}$, and the random seed, every allocation decision is uniquely determined and exactly reproducible: the MICP formulation solved with MOSEK (MOSEK ApS, 2024) guarantees global optimality by construction, verified across all 912 optimization runs, while the position bounds and net exposure limit are the sole levers through which risk tolerance enters the strategy. No multiple restarts, initialization choices, or ex-post corrections are required or applied.

The main result is that the strategy continues to generate substantial risk-adjusted returns in a market environment quite different from the original sample. Under $\bar{w} = 2\%$ and $\delta_{\max} = 0.05$, the OOPS earns a Sharpe ratio of 1.18, a certainty equivalent of 17.04%, and an approximately symmetric return

distribution (skewness +0.09), with roughly 101% of wealth in Treasury bills. Factor regressions show that the market excess return is the only significant contemporaneous factor ($\beta = 0.259$, $p = 0.007$). No lagged variable significantly predicts returns. The remaining return variation is unexplained by the factors we consider.

The sensitivity analysis reveals that these results depend critically on tight position constraints. The CE optimum is sharply localized around $\bar{w} \in [2\%, 5\%]$ and $\delta_{\max} \in [5\%, 10\%]$; beyond $\bar{w} = 7\%$, performance collapses as single adverse months destroy CRRA utility. Positive skewness—the feature that distinguishes the OOPS from simple short option strategies—exists only for $\bar{w} \leq 3\%$. These constraints also serve as the primary safeguard against scenario degeneracy in the bootstrap, a point discussed in detail in Section 4.5.

Several limitations, documented in Section 4.5, qualify these findings. The sample period is dominated by a prolonged bull market that structurally favors option-writing strategies, the 152-month sample lacks statistical power to distinguish crash-risk compensation from mispricing alpha, and the bootstrap’s reliance on short volatility windows can produce overly compressed scenario distributions. How the strategy would perform in a sustained bear market or high-volatility regime remains an open question for future research. By providing a fully transparent and reproducible implementation that relies exclusively on accessible resources—MOSEK is available under a free academic license, and the complete options dataset was obtained from Databento at negligible cost—we hope to lower the barrier to entry for researchers without access to proprietary databases such as OptionMetrics, and to enable further investigation into this area of research.

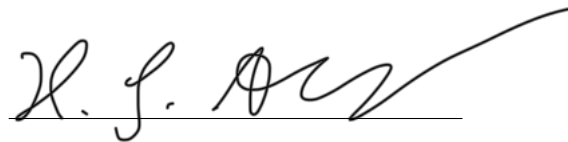
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Freiburg im Breisgau, den 26. Februar 2026

A handwritten signature in black ink, consisting of stylized cursive letters, positioned above a horizontal line.

Houshyar Amini